

COMSATS University Islamabad

Attock Campus



Department Of Computer Science

<i>Course</i>	<i>Information security lab</i>
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Student Details

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Task: 1

Write a Python function to calculate GCD using Euclid's Algorithm

```
def gcd_euclid(a, b):  
    while b != 0:  
        a, b = b, a % b  
    return a
```

Example

```
print("GCD:", gcd_euclid(48, 18))
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\PythonPro  
GCD: 6  
  
Process finished with exit code 0
```

Task: 2

GCD with Step Display Modify the GCD function to display each step of Euclid's Algorithm.

```
def gcd_steps(a, b):  
    print("Steps of Euclid's Algorithm:")  
    while b != 0:  
        print(f"{a} = {b} * ({a//b}) + {a%b}")  
        a, b = b, a % b  
    print("GCD =", a)  
    return a
```

```
gcd_steps(48, 18)
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\PythonPro  
Steps of Euclid's Algorithm:  
48 = 18 * (2) + 12  
18 = 12 * (1) + 6  
12 = 6 * (2) + 0  
GCD = 6  
  
Process finished with exit code 0
```

Task: 3

Recursive Implementation (5 Marks)

Implement Euclid's Algorithm using recursion instead of a loop.

```
def gcd_recursive(a, b):  
    if b == 0:  
        return a  
    return gcd_recursive(b, a % b)  
  
print("GCD (Recursive):", gcd_recursive(48, 18))
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\PythonProje  
GCD (Recursive): 6  
  
Process finished with exit code 0
```